A Brauer–Manin obstruction to integral points

Worked out on the negative Pell conic $x^2 - 34y^2 = -1$: rational points are dense, local integral points exist everywhere, and a single quaternion class kills every integral point

every claim below is checked in `integral-bm.gp`*; output in* `results/integral-bm.txt`

The example in one line. Let $\mathcal{X} \subset \mathbb{A}^2_{\mathbb{Z}}$ be

$$\mathcal{X}: \quad x^2 - 34y^2 = -1,$$

and let $\mathcal{A} = (2, x+4)$ be the quaternion algebra over $\mathbb{Q}(\mathcal{X})$. Then $\mathcal{X}(\mathbb{Z}_v) \neq \varnothing$ for every place v, and $\mathcal{X}(\mathbb{Q})$ is infinite and Zariski dense — but

$$\sum_v \mathrm{inv}_v \mathcal{A}(P_v) = \frac{1}{2} \quad \text{for every} \quad (P_v) \in \mathcal{X}(\boldsymbol{A}_{\mathbb{Z}}),$$

so $\mathcal{X}(\boldsymbol{A}_{\mathbb{Z}})^{\mathcal{A}} = \varnothing$ and hence $\mathcal{X}(\mathbb{Z}) = \varnothing$. The entire obstruction sits at the prime 2: at every $\mathbb{Z}_2$-point $x \equiv \pm 1 \bmod 8$, which makes $x + 4 \equiv \pm 3 \bmod 8$, which makes 2 a non-norm.

1. What the integral obstruction is, and how it differs from the rational one

Let $\mathcal{X}$ be a separated scheme of finite type over $\mathbb{Z}$ with generic fibre $X = \mathcal{X}_{\mathbb{Q}}$. The **integral adelic points** are

$$\mathcal{X}(\boldsymbol{A}_{\mathbb{Z}}) = \mathcal{X}(\mathbb{R}) \times \prod_p \mathcal{X}(\mathbb{Z}_p) \subseteq X(\boldsymbol{A}_{\mathbb{Q}}),$$

a **closed** subset of the rational adelic points, and a much smaller one: it is the locus where every coordinate is a p-adic integer at every p. For $\mathcal{B} \in \mathrm{Br}(X)$ the evaluation $P \mapsto \mathrm{inv}_v \mathcal{B}(P_v) \in \mathbb{Q}/\mathbb{Z}$ is locally constant on $X(\mathbb{Q}_v)$ and vanishes for almost all v, and one sets

$$\mathcal{X}(\boldsymbol{A}_{\mathbb{Z}})^{\mathrm{Br}} = \left\{ (P_v) : \sum_v \mathrm{inv}_v \mathcal{B}(P_v) = 0 \ \text{ for all } \ \mathcal{B} \right\}.$$

Because a global point $P \in \mathcal{X}(\mathbb{Z}) \subseteq X(\mathbb{Q})$ has $\mathrm{inv}_v \mathcal{B}(P) = \mathrm{inv}_v$ of a single class in $\mathrm{Br}(\mathbb{Q})$, class field theory ($\sum_v \mathrm{inv}_v = 0$ on $\mathrm{Br}(\mathbb{Q})$) gives

$$\mathcal{X}(\mathbb{Z}) \subseteq \mathcal{X}(\boldsymbol{A}_{\mathbb{Z}})^{\mathrm{Br}} \subseteq \mathcal{X}(\boldsymbol{A}_{\mathbb{Z}}).$$

An emptiness of the middle set with non-emptiness of the right one is a **Brauer–Manin obstruction to an integral point**.

Why this can bite when the rational obstruction does not. The pairing is the same pairing; only the domain shrinks. A class $\mathcal{B}$ may take both values 0 and $1/2$ on $X(\mathbb{Q}_p)$ but only one of them on the compact subset $\mathcal{X}(\mathbb{Z}_p)$. That is exactly what happens below: the invariant at an odd p is $\chi(2)^{v_p(x+4)}$, which is non-trivial only when x has a **pole** at p — something a rational point may afford and an integral point may not. So the mechanism is not exotic: **the class measures denominators**, and integrality is the statement that there are none.

Two remarks on the definition, since both matter here.

- One may pair against $\mathrm{Br}(\mathcal{X})$ (the $\mathbb{Z}$-scheme) or against the larger $\mathrm{Br}(X)$ of the generic fibre; the latter gives a possibly finer obstruction and is what is used below. All the argument needs is that $\mathcal{A}$ lies in

$\mathrm{Br}(X)$, so that its invariants are defined at every point of $\mathcal{X}(\boldsymbol{A}_{\mathbb{Z}}) \subseteq X(\boldsymbol{A}_{\mathbb{Q}})$ and reciprocity applies to the rational ones.

- The integral analogue of the Hasse principle is **strong** approximation, not weak; the obstruction below is an obstruction to strong approximation on the group $\boldsymbol{G}_m^1$-torsor $\mathcal{X}$, and by a theorem of Colliot-Thélène and Xu for homogeneous spaces of connected groups it is the only one. So $\mathcal{X}(\mathbb{Z}) = \varnothing$ is not merely implied by the computation — the computation is the reason.

2. The variety

Take $\mathcal{X} : x^2 - 34y^2 = -1$ over $\mathbb{Z}$; write $K = \mathbb{Q}(\sqrt{34})$, so $\mathcal{X}$ is the affine conic $N_{K/\mathbb{Q}}(x + y\sqrt{34}) = -1$, a torsor under the norm-one torus $T = R^1_{K/\mathbb{Q}}\boldsymbol{G}_m$.

2.1. No integral point

$\mathcal{O}_K = \mathbb{Z}[\sqrt{34}]$ and its fundamental unit is $35 + 6\sqrt{34}$, of norm $+1$. An integral point is a unit of norm -1, so there is none: $\boldsymbol{\mathcal{X}(\mathbb{Z}) = \varnothing}$**.** (Checked also by brute force for $0 \le y \le 2 \cdot 10^5$.)

2.2. Local integral points everywhere

At $v = \infty$: $y = 1$, $x = \sqrt{33}$.

At $v = 2$: again $y = 1$, so $x^2 = 33$; and $33 \equiv 1 \bmod 8$, so 33 is a square in $\mathbb{Z}_2$.

At odd p: the gradient $(2x, -68y)$ of $x^2 - 34y^2 + 1$ vanishes mod p only if $x \equiv 0$ and $34y \equiv 0$, which contradicts the equation; so $\mathcal{X}$ is smooth over $\mathbb{Z}[1/2]$ and Hensel lifts $\mathbb{F}_p$-points. A point mod p exists for every odd p (verified for all $p < 2000$; in general the affine conic over $\mathbb{F}_p$ has $p \pm 1 \ge 2$ points). So $\boldsymbol{\mathcal{X}(\mathbb{Z}_v) \neq \varnothing}$ **for all** $\boldsymbol{v}$**.**

2.3. Rational points are dense

$(3/5, 1/5)$ is a point: $9/25 - 34/25 = -1$. So $X \cong \mathbb{P}^1_{\mathbb{Q}}$ minus one closed point P_∞ of degree 2 (the locus $x = \pm\sqrt{34}y$ at infinity, with residue field K), and $X(\mathbb{Q})$ is infinite and Zariski dense. **This example is therefore about integrality and nothing else.**

3. The Brauer class

We want a non-constant class in $\mathrm{Br}(X)$. Since $X = \mathbb{P}^1_{\mathbb{Q}} \setminus P_\infty$, the residue sequence

$$0 \longrightarrow \mathrm{Br}(\mathbb{P}^1_{\mathbb{Q}}) \longrightarrow \mathrm{Br}(\mathbb{Q}(X)) \xrightarrow{\oplus\partial_Q} \oplus_{Q \in (\mathbb{P}^1)^{(1)}} H^1(\kappa(Q), \mathbb{Q}/\mathbb{Z})$$

says that a class lies in $\mathrm{Br}(X)$ exactly when all its residues **on** $\boldsymbol{X}$ vanish; a residue at the deleted point P_∞ is allowed, and a non-zero one there makes the class non-constant.

Try $\mathcal{A} = (2, \lambda)$ with $\lambda = x + c$, $c \in \mathbb{Z}$. For a constant a the tame symbol is $\partial_Q(a, f) = \overline{a}^{v_Q(f)} \in \kappa(Q)^\times / \kappa(Q)^{\times 2}$, so:

- **At** $\boldsymbol{P_\infty}$, $v(x + c) = -1$ and $\kappa(P_\infty) = K = \mathbb{Q}(\sqrt{34})$. Since 2 is not a square in K, the residue is non-trivial and $\mathcal{A}$ is non-constant.
- **At the zeros of** $\boldsymbol{x + c}$, which form the closed subscheme $x = -c$, $34y^2 = c^2 + 1$: the residue field is $\mathbb{Q}(\sqrt{34(c^2+1)})$, and the residue is the class of 2 there. It is trivial precisely when 2 is a square in that field, i.e. when $34(c^2 + 1) \in 2(\mathbb{Q}^\times)^2$, i.e. when

$$17(c^2 + 1) \text{ is a perfect square.}$$

The choice $\boldsymbol{c = 4}$**.** $17 \cdot (4^2 + 1) = 17 \cdot 17 = 289 = 17^2$. So the zeros of $x + 4$ on X are $x = -4$, $y = \pm 1/\sqrt{2}$, a single closed point with residue field $\mathbb{Q}(\sqrt{2})$ — and 2 **is** a square there. Hence

$$\mathcal{A} = (2, x + 4) \in \mathrm{Br}(X), \quad \mathcal{A} \notin \mathrm{Br}(\mathbb{Q}).$$

There is a second representative. On X,

$$(2, x+4)(2, x-4) = (2, x^2-16) = (2, 34y^2-17) = (2, 17)(2, 2y^2-1),$$

and $(2,17)$ splits because $17 \equiv 1 \bmod 8$, while $2y^2 - 1 = -N_{\mathbb{Q}(\sqrt{2})/\mathbb{Q}}\big(1 + y\sqrt{2}\big)$ gives $(2, 2y^2-1) = (2, -1) = 1$. So $(2, x+4) = (2, x-4)$, and one of the two is regular at any given point — useful at the point $x = -4$ itself.

4. The evaluation, place by place

Throughout, $\mathrm{inv}_v \mathcal{A}(P) = 1/2$ iff the Hilbert symbol $(2, x+4)_v = -1$.

4.1. The real place: always 0

$(a, b)_\infty = -1$ requires $a < 0$ **and** $b < 0$, and here $a = 2 > 0$. So $\mathrm{inv}_\infty \mathcal{A} = 0$ at every real point, whatever the sign of $x + 4$.

4.2. The prime 2: always 1/2

Proposition. For every $(x, y) \in \mathcal{X}(\mathbb{Z}_2)$ one has $x \equiv \pm 1 \bmod 8$, and therefore $\mathrm{inv}_2 \mathcal{A} = 1/2$.

Proof. Write the equation as $x^2 + 1 = 34y^2 = 2 \cdot 17y^2$. The right side has odd 2-valuation $1 + 2v_2(y)$. If x were even, $x^2 + 1$ would be a unit; so x is odd, hence $x^2 \equiv 1 \bmod 8$, hence $v_2(x^2+1) = 1$, hence $v_2(y) = 0$. Now divide by 2:

$$(x^2+1)/2 = 17y^2 \equiv 17 \equiv 1 \bmod 8,$$

using $y^2 \equiv 1 \bmod 8$ for a 2-adic unit y. But if $x \equiv \pm 3 \bmod 8$ then $x^2 \equiv 9 \bmod 16$ and $(x^2+1)/2 \equiv 5 \bmod 8$. Contradiction; so $x \equiv \pm 1 \bmod 8$.

Then $x + 4 \equiv 5$ or $3 \bmod 8$, i.e. $x + 4 \equiv \pm 3 \bmod 8$, and for an odd 2-adic unit u one has $(2, u)_2 = +1$ iff $u \equiv \pm 1 \bmod 8$. So $(2, x+4)_2 = -1$. ∎

An exhaustive machine check over $\mathbb{Z}/2^{10}$ finds 2048 solutions, all with $x \bmod 8 \in \{1, 7\}$ and all with $(2, x+4)_2 = -1$.

4.3. Odd primes: always 0

Proposition. For every odd p and every $(x, y) \in \mathcal{X}(\mathbb{Z}_p)$, $\mathrm{inv}_p \mathcal{A} = 0$.

Proof. For odd p and a constant a, $(a, f)_p = ((a/p))^{v_p(f)}$. So $(2, x+4)_p = -1$ forces both $(2/p) = -1$ (i.e. $p \equiv \pm 3 \bmod 8$) and $v_p(x+4)$ odd, in particular $p \mid x + 4$.

Suppose $p \mid x + 4$, so $x \equiv -4$. Then $34y^2 = x^2 + 1 \equiv 17 \bmod p$. Here $p \neq 17$, because $17 \equiv 1 \bmod 8$ has $(2/17) = +1$; so 17 is invertible and $2y^2 \equiv 1 \bmod p$, with y a unit. Thus $2 \equiv \left(y^{-1}\right)^2$ is a square mod p, contradicting $(2/p) = -1$. ∎

(The two conditions "$p \mid x + 4$ is realisable on $\mathcal{X}(\mathbb{Z}_p)$" and "2 is a square mod p" were also compared directly for all odd $p < 5000$: they agree at every prime.)

4.4. The sum

Theorem. For every $(P_v) \in \mathcal{X}(\boldsymbol{A}_\mathbb{Z})$,

$$\sum_v \mathrm{inv}_v \mathcal{A}(P_v) = \underbrace{0}_{v=\infty} + \underbrace{1/2}_{v=2} + \underbrace{0}_{v \text{ odd}} = \frac{1}{2} \neq 0.$$

Hence $\mathcal{X}(\boldsymbol{A}_\mathbb{Z})^\mathcal{A} = \varnothing$ while $\mathcal{X}(\boldsymbol{A}_\mathbb{Z}) \neq \varnothing$: a Brauer–Manin obstruction to an integral point, on a variety with a dense set of rational points.

5. How the rational points escape

This is the interesting half, and it is what makes the example an **integral** one. Reciprocity forces $\sum_v \mathrm{inv}_v \mathcal{A}(P) = 0$ at every $P \in X(\mathbb{Q})$, and the theorem says $\mathrm{inv}_2 = 1/2$ whenever P is 2-integral and $\mathrm{inv}_p = 0$ whenever P is p-integral. So **every rational point must fail to be p-integral at some odd $p \equiv \pm 3 \bmod 8$**, and the failure must be visible as an odd pole order of $x+4$, i.e. as a denominator.

Here are all rational points of denominator ≤ 45, with the places where the invariant is $1/2$:

x	y	denom	places with inv $= 1/2$	that odd p mod 8
$5/3$	$1/3$	3	$\{2,3\}$	3
$29/3$	$5/3$	3	$\{2,3\}$	3
$3/5$	$1/5$	5	$\{2,5\}$	5
$99/5$	$17/5$	5	$\{2,5\}$	5
$27/11$	$5/11$	11	$\{2,11\}$	3
$75/11$	$13/11$	11	$\{2,11\}$	3
$11/27$	$5/27$	27	$\{2,3\}$	3
$3/29$	$5/29$	29	$\{2,29\}$	5
$141/37$	$25/37$	37	$\{2,37\}$	5
$61/45$	$13/45$	45	$\{2,5\}$	5

Every row has an even number of bad places — Hilbert reciprocity — and every row contains 2 together with exactly one odd prime, which is $\equiv \pm 3 \bmod 8$ and divides the denominator. Nothing here is a coincidence: it is the two propositions of Section 4 read backwards.

The same holds along the orbit of $(3/5, 1/5)$ under multiplication by the fundamental unit $35 + 6\sqrt{34}$:

$$\left(\frac{3}{5}, \frac{1}{5}\right) \longrightarrow \left(\frac{309}{5}, \frac{53}{5}\right) \longrightarrow \left(\frac{21627}{5}, \frac{3709}{5}\right) \longrightarrow \left(\frac{1513581}{5}, \frac{259577}{5}\right) \longrightarrow \dots,$$

all with denominator 5, all with $\mathrm{inv}_5 = 1/2$. The denominator 5 never goes away, and $5 \equiv 5 \bmod 8$. The rational points are dense, and **every single one of them carries a pole at a prime where 2 is a non-residue.**

> **The escape hatch, directly.** Take $p = 3$ and a $\mathbb{Q}_3$-point with $v_3(x) = -1$, say $x = a/3$, $y = b/3$ with a, b units: the equation becomes $a^2 + 9 = 34b^2$, which mod 9 asks that 7 be a square — and $4^2 = 16 \equiv 7 \bmod 9$. Such points exist, they have $v_3(x+4) = -1$ odd, and $\mathrm{inv}_3 = 1/2$. They are not 3-integral. The class $\mathcal{A}$ takes both values on $X(\mathbb{Q}_3)$ and only one value on $\mathcal{X}(\mathbb{Z}_3)$ — the phenomenon described in Section 1, with numbers.

6. What the class really is: genus theory

The residue of $\mathcal{A}$ at P_∞ is the class of 2 in $K^\times/K^{\times 2}$, $K = \mathbb{Q}\left(\sqrt{34}\right)$ — that is, the quadratic character of K cutting out

$$K\left(\sqrt{2}\right) = \mathbb{Q}\left(\sqrt{2}, \sqrt{17}\right),$$

which is **the genus field of K**: the unramified quadratic extension of K corresponding to the factorisation $34 = 2 \cdot 17$. PARI gives $h(K) = 2$ with class group $\mathbb{Z}/2$, generated by that genus class.

So the Brauer–Manin obstruction here is not new mathematics dressed up — it **is** the classical genus-theory obstruction to the negative Pell equation, written as a sum of local invariants. Two consistency checks that it had to be this class:

- The residue must be **unramified everywhere** over K, or $\mathcal{A}$ would not extend over the integral model in the way the argument uses; the unramified quadratic characters of K are the genus characters.
- The residue must lie in $\ker\left(\mathrm{cor}_{K/\mathbb{Q}}\right)$ for the class to descend, and corestriction on square classes is the norm: $N_{K/\mathbb{Q}}(2) = 4$, a square. ✓

This also explains the shape of the answer. $\mathrm{Br}(\mathcal{X})/\mathrm{Br}(\mathbb{Z}) \cong \mathbb{Z}/2$ for a torsor under $T = R^1_{K/\mathbb{Q}}\boldsymbol{G}_m$, since $H^1\left(\mathbb{Q}, \hat{T}\right) = \hat{H}^1(\mathbb{Z}/2, \mathbb{Z}^-) = \mathbb{Z}/2$; there is one class to try, and it works.

7. Is 34 special?

No — and it is worth checking, because a single example proves nothing about the method. Run the same recipe for $d = 2q$ with $q \equiv 1 \bmod 8$ prime: find c with $q(c^2+1)$ a square (a point on an auxiliary Pell conic), form $\mathcal{A} = (2, x+c)$ on $x^2 - dy^2 = -1$, and compute inv_2 over all of $\mathbb{Z}/2^{10}$.

d	q	c	inv_2 on $\mathcal{X}(\mathbb{Z}_2)$	negative Pell
34	17	4	1/2 (constant)	insoluble
82	41	32	0 (constant)	soluble
146	73	1068	1/2 (constant)	insoluble
178	89	500	1/2 (constant)	insoluble
194	97	5604	1/2 (constant)	insoluble
226	113	776	0 (constant)	soluble
274	137	1744	0 (constant)	soluble

The class detects the obstruction exactly when there is one, and is silent exactly when there is not: seven for seven. In each case the residue field of the zeros of $x+c$ is $\mathbb{Q}\left(\sqrt{2}\right)$, so the class is unramified on X by the same computation as for $d = 34$; and $d = 34$ is the smallest such d.

8. Summary

On $\mathcal{X} : x^2 - 34y^2 = -1$ over $\mathbb{Z}$ the quaternion class $\mathcal{A} = (2, x+4)$ is unramified on the generic fibre, non-constant, and evaluates to $1/2$ at $v = 2$ and to 0 at every other place, **for every integral adelic point**. Hence $\mathcal{X}(\boldsymbol{A}_\mathbb{Z}) \neq \varnothing$, $\mathcal{X}(\boldsymbol{A}_\mathbb{Z})^{\mathrm{Br}} = \varnothing$, $\mathcal{X}(\mathbb{Z}) = \varnothing$. Rational points are dense and satisfy $\sum_v \mathrm{inv}_v = 0$ by reciprocity; they manage it by having a denominator at a prime $p \equiv \pm 3 \bmod 8$, and that denominator is precisely what an integral point is forbidden. The class is the genus character of $\mathbb{Q}\left(\sqrt{34}\right)$, so the obstruction is the classical one — but the Brauer formulation is the one that generalises, and by Colliot-Thélène–Xu it is, for such torsors, the only obstruction there is.

Companion file: `integral-bm.gp`, run as

```
gp -q -s 2000000000 integral-bm.gp < /dev/null > results/integral-bm.txt
```

It verifies (1) the norm of the fundamental unit, (2) local solubility at every place, (3) the residue computation for $\mathcal{A}$, (4)–(6) the three invariant computations, (8) reciprocity and the denominators of rational points, (9) the genus-field identification, and (10) the family scan of Section 7.